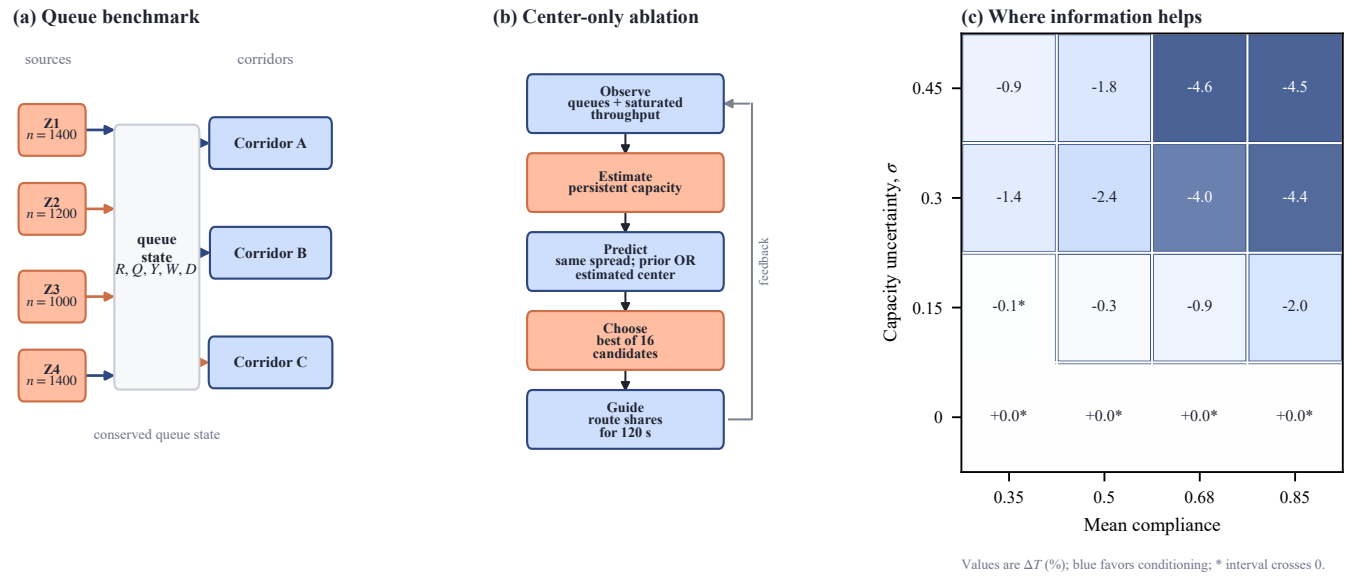

(a) Queue benchmark
sources
corridors
Z1
n = 1400
Z2
n = 1200
Z3
n = 1000
Z4
n = 1400
queue
state
R, Q, Y, W, D
Corridor A
Corridor B
Corridor C
conserved queue state
(b) Center-only ablation
Observe
queues + saturated
throughput
Estimate
persistent capacity
Predict
same spread; prior OR
estimated center
Choose
best of 16
candidates
Guide
route shares
for 120 s
feedback
(c) Where information helps
Capacity uncertainty, σ
0.45
0.3
0.15
0
-0.9
-1.8
-4.6
-4.5
-1.4
-2.4
-4.0
-4.4
-0.1*
-0.3
-0.9
-2.0
+0.0*
+0.0*
+0.0*
+0.0*
0.35
0.5
0.68
0.85
Mean compliance
Values are ΔT (%); blue favors conditioning; * interval crosses 0.